REPORT 2: UML CLASS DIAGRAM REPORT

Project Title: Hospital Management System

. Course: Object-Oriented Analysis and Design / UML Modeling

1. System Overview

The UML Class Diagram for the Hospital Management System models the static structure of the proposed system. It illustrates the system's core entities, their internal attributes and methods, and the structural relationships connecting them.

2. Class Specifications & Structures

2.1 Class: Patient

- Description: Represents individuals receiving medical services.
- Attributes:
  - PatientID : String — Unique identifier for the patient.
  - FullName : String — Full legal name of the patient.
  - age : int — Age of the patient.
  - gender : string — Gender of the patient.
  - PhoneNumber : string — Contact phone number

Methods:

- + register() — Registers a new patient into the system.
- + updateprofile() — Updates existing patient profile details.
- + bookAppointment() — Initiates an appointment request.

2.2 Class: Appointment

- Description: Manages scheduled consultation interactions.
- **Attributes:**

- appointmentID : String — Unique identifier for the appointment.
- appointmentData : String — Date and time for the scheduled visit.
- status : string — Current state of the appointment (e.g., Pending, Confirmed, Canceled).

Methods:

- + confirmAppointment() — Confirms the scheduled appointment slot.
- + cancelAppointment() — Cancels the scheduled appointment.

2.3 Class: Doctor

- Description: Represents medical specialists conducting consultations.
- **Attributes:**
  - DoctorId : string — Unique identifier for the doctor.
  - FullName : String — Full legal name of the doctor.
  - Specialization : string — Area of medical expertise (e.g., Pediatrics, Neurology).
  - email : string — Work email address.

Methods:

- + viewSchedule() — Displays assigned daily appointment slots.
- + Diagnosepatient() — Records medical diagnoses for patients.
- + prescribeMedication() — Issues medication prescriptions.

2.4 Class: Department

- Description: Represents organizational divisions within the hospital.
- **Attributes:**
- deptID : String — Unique identifier for the department.
- deptName : String — Name of the medical division.

Methods:

- + addDoctor() — Assigns a doctor to the department.
- + getDepartmentDetails() — Retrieves information about the department.

2.5 Class: Bill

- Description: Handles financial transactions and billing invoices.
- **Attributes:**
- billID : String — Unique invoice identifier.
- totalAmount : double — Total financial cost of services rendered.
- paymentStatus : string — Current payment condition (e.g., Paid, Unpaid).

Methods:

- + generateBill() — Calculates and creates a bill invoice.
- + processpayment() — Updates payment status upon receiving payment.

3. Class Relationships and Multiplicities

3.1 Patient to Appointment

- Relationship Type: Association
- Multiplicity: 1 to * (1 : *)

- Explanation: One Patient can schedule multiple Appointments over time (*), but each Appointment is associated with exactly one (1) Patient.

3.2 Doctor to Appointment

- Relationship Type: Association
- Multiplicity: 1 to * (1 : *)
- Explanation: One Doctor can be assigned to multiple Appointments (*) in their schedule, while each Appointment is assigned to exactly one (1) Doctor.

3.3 Department to Doctor

- Relationship Type: Aggregation
- Multiplicity: 1 to * (1 : *)
- Explanation: One Department consists of multiple (*) Doctors working within that specialty. Each Doctor belongs to one (1) primary Department.

3.4 Patient to Bill

- Relationship Type: Association
- Multiplicity: 1 to * (1 : *)
- Explanation: One Patient can accumulate multiple (*) Bills across different hospital visits, whereas each Bill is generated for exactly one (1) Patient.